

From Handcrafted Features to Transformers: A Comparative Study of Music Genre Classification

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Abstract

This project presents a comprehensive comparative study of music genre classification using classical machine learning, deep learning, and transformer-based approaches on the GTZAN dataset. We evaluate handcrafted feature-based models such as Logistic Regression and Random Forest, deep learning models using Mel-spectrograms with Convolutional Neural Networks (CNNs), including a transfer learning-based ResNet18 model, and a hybrid CNN-LSTM architecture. Additionally, pretrained representation learning models such as Wav2Vec2 and Music Embedding Representation Transformer (MERT) are explored for learning features directly from raw audio.

The results show that classical models provide strong baseline performance, while deep learning models significantly improve accuracy by learning hierarchical time-frequency representations. The hybrid CNN-LSTM model further enhances performance by capturing both spatial and temporal features. In contrast, Wav2Vec2 performs poorly in this task due to its focus on speech data. Among all approaches, the MERT-based model achieves the best performance, demonstrating the effectiveness of domain-specific pretrained representations for music classification.

1 Introduction

Music Genre Classification (MGC) is an important problem in Music Information Retrieval (MIR), with applications in recommendation systems, playlist generation, music organization, and streaming services such as Spotify and Apple Music. As the volume of digital music grows exponentially, manual classification becomes infeasible, making automated systems essential.

Music signals are inherently complex, consisting of multiple overlapping components such as melody, harmony, rhythm, and timbre. Unlike speech signals, music does not follow a strict structure, making classification more challenging. Different genres may share similar acoustic patterns, leading to ambiguity in classification.

Traditional approaches rely on handcrafted feature extraction techniques such as Mel-Frequency Cepstral Coefficients (MFCCs), spectral centroid, chroma features, and

zero-crossing rate. These features are then used with machine learning algorithms like Logistic Regression, Decision Trees, and K-Nearest Neighbors.

However, these approaches have limitations in capturing long-term dependencies and complex hierarchical patterns in audio signals. Recent advancements in deep learning have enabled end-to-end learning from raw audio or spectrogram representations. Convolutional Neural Networks (CNNs) have shown strong performance in capturing spatial patterns in spectrograms, while transformer-based models such as MERT leverage large-scale pretraining to learn high-level representations.

In this project, we aim to:

- Compare classical machine learning models with deep learning approaches and transformers.
- Analyze the impact of feature engineering and dimensionality reduction.
- Study the effect of temporal segmentation (30-sec vs 3-sec).
- Evaluate performance using multiple metrics and visualization techniques.

2 Related Work

One of the earliest works by *Tzanetakis and Cook* (2002) introduced the GTZAN dataset and used handcrafted audio features for classification, achieving about 60%–65% accuracy with classical models.

Later, researchers applied traditional machine learning methods such as k-NN, SVM, and Random Forest using features like MFCCs, chroma, and spectral descriptors. These approaches typically achieved accuracies between 70% and 80%, depending on feature quality, preprocessing, and validation methodology.

With the rise of deep learning, Convolutional Neural Networks (CNNs) became popular for this task. By using Mel-spectrograms as input, CNNs can automatically learn useful patterns from audio. Models like CRNN proposed by *Choi et al.* (2016), along with more recent CNN-based architectures, improved performance to around 85%–90% on the GTZAN dataset.

Recently, self-supervised models such as **Wav2Vec2** have been used to learn features directly from raw audio. While these models perform very well in speech tasks, their performance on music classification is generally lower unless fine-tuned specifically for music data. To improve this, newer models like **MERT** are trained specifically on music datasets. These models are better at capturing musical properties such as rhythm and harmony, and recent studies report strong performance on music classification tasks.

3 Dataset Description

The dataset used in this study is the GTZAN dataset, organized into the following components:

- **genres_original/** Contains raw audio files categorized into 10 genres (100 songs each, 30s duration).
- **images_original/** Contains spectrogram images generated from audio signals.
- **features_30_sec.csv** Features extracted from full 30-second tracks. Shape: (1000, 60).
- **features_3_sec.csv** Features extracted from 3-second segments. Shape: (9990, 60).

4 Project Workflow

To prevent formatting overlaps, the system architecture is defined via three parallel pipelines (see Fig. 2):

1. **Preprocessing Phase:** GTZAN audio (30s and 3s) is segmented and normalized. Handcrafted features undergo standard scaling.
2. **Classical ML Pipeline:** Logistic Regression, Random Forest, and a Weighted Hybrid model are trained on features. PCA is applied to retain 95% variance.
3. **Deep Learning Pipeline:** 128x128 Mel-spectrograms are fed into a custom CNN, a pretrained ResNet18, and a novel CNN-LSTM architecture.
4. **Transformer Pipeline:** Raw 24kHz audio is passed through Wav2Vec2 and MERT for embedding extraction, followed by an MLP classifier.
5. **Evaluation & Output:** Models are evaluated using Macro F1 and Accuracy. The final embeddings power a content-based Music Recommendation System.

5 Implementation Summary

The proposed framework combines classical machine learning, deep learning, and pretrained transformer-based methods to enable a structured comparison. The dataset is split into **70% training, 15% validation, and 15% testing sets** to ensure reliable evaluation. Table 1 summarizes the key implementation details across the three pipelines, demonstrating the evolution from manually extracted parameters to learned embeddings.

6 TASK 1. Feature-based Classification

We utilize pre-extracted features from the GTZAN dataset. These features capture key spectral and temporal properties, including MFCCs, chroma features, spectral descriptors, and RMS energy.

Feature importance analysis highlights key contributors such as **percept_r_var**, **chroma_stft_mean**, **mfcc4_mean**, and **rms_mean**.

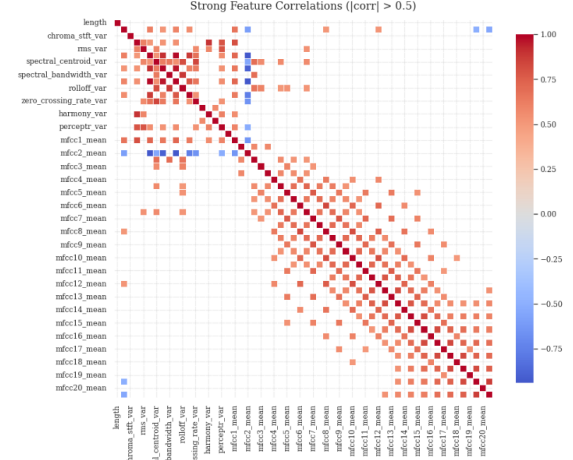


Figure 4: Feature correlation matrix.

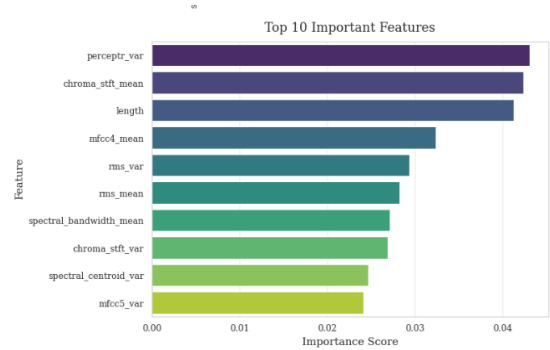


Figure 5: Feature importance using Random Forest.

6.1 Data Leakage Prevention

During initial experiments on the 3-second dataset, we observed unexpectedly high classification performance. Upon closer inspection, we identified **data leakage** caused by a naive random split, where segments originating from the same parent track were distributed across both training and testing sets. To mitigate this, we enforced a **strict track-level split**.

6.2 Weighted Hybrid Model (Novelty)

We propose a **Weighted Hybrid Model** that combines the strengths of complementary classifiers: **Random Forest**

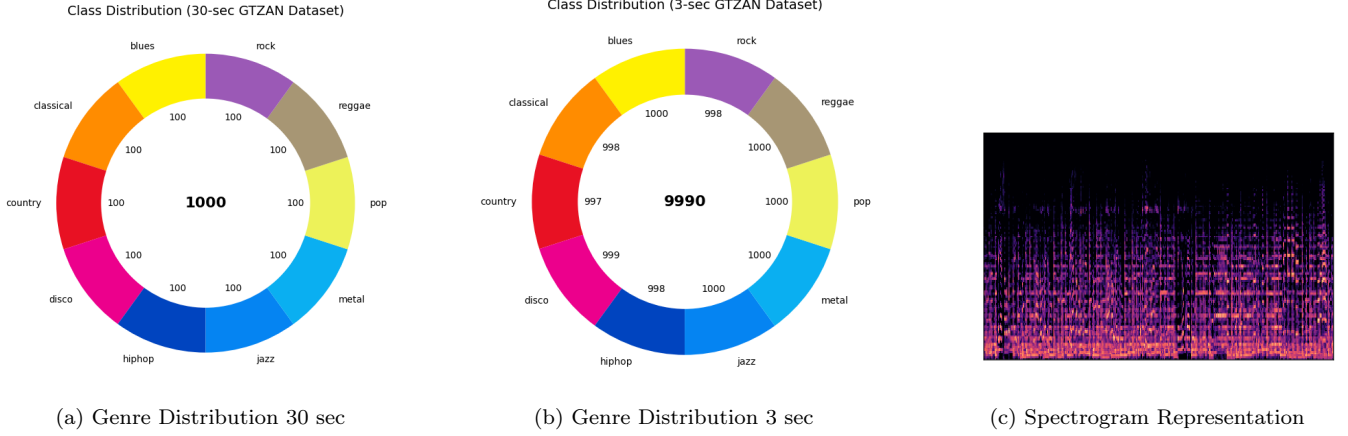


Figure 1: Visualization of the GTZAN dataset showing the class distribution for both 30-second and 3-second segments, and spectrogram representations. The spectrogram shown corresponds to the audio sample `blues.00002`.

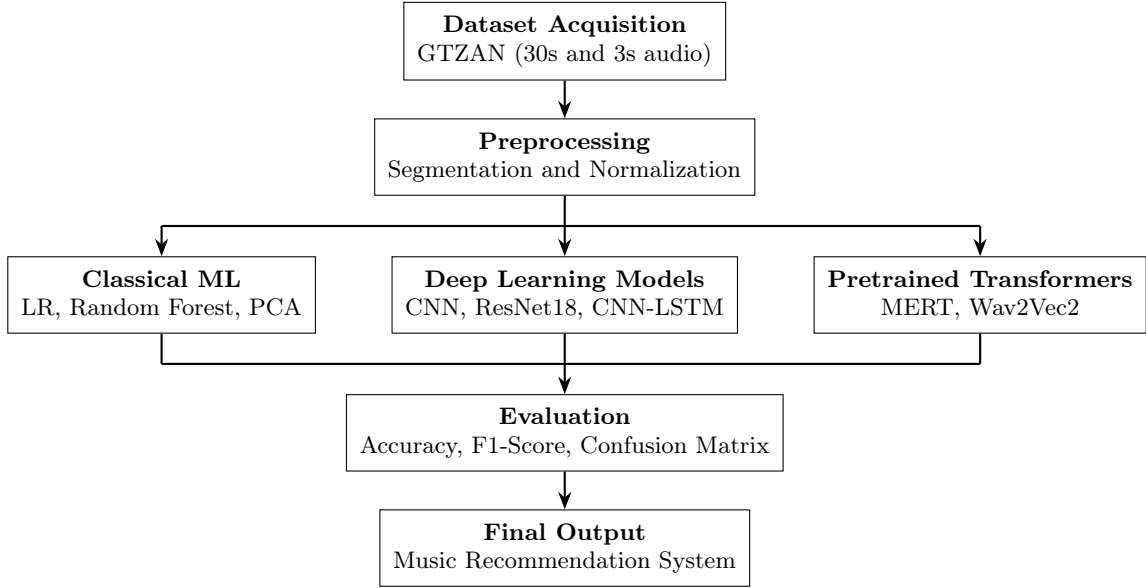


Figure 2: Proposed System Workflow for Music Genre Classification.

(**RF**) and **Logistic Regression (LR)**. This ensemble fuses the class probability outputs using a weighted sum:

$$P_{\text{final}} = \alpha \cdot P_{\text{RF}} + (1 - \alpha) \cdot P_{\text{LR}} \quad (1)$$

where $\alpha \in [0, 1]$ controls the contribution. For the 3-second dataset, predictions are aggregated at the track level using **majority voting**. The proposed hybrid framework strategically integrates the non-linear decision-making capability of Random Forest with the simplicity and strong generalization behavior of Logistic Regression. By combining probabilistic outputs instead of hard predictions, the model captures complementary decision boundaries and produces more reliable genre predictions. This lightweight yet effective ensemble significantly improves robustness against noisy audio segments while maintaining computational efficiency. Experimental results demonstrate that the proposed method consistently outperforms individual classical machine learning baselines,

highlighting the effectiveness of probability-level fusion for music genre classification.

Table 2: Classical Model Performance

Configuration	Dataset	Acc	F1
Logistic Regression	30-sec	0.710	0.711
Random Forest	30-sec	0.690	0.682
Hybrid (RF+LR)	30-sec	0.745	0.745
Logistic Regression	3-sec	0.760	0.754
Random Forest	3-sec	0.775	0.769
Hybrid (RF+LR)	3-sec	0.810	0.807

6.3 Discussion and Insights

Effect of Model Selection: The choice of classifier has a strong impact on genre classification performance. Logistic

Regression (LR) performs better on the smaller 30-sec dataset because simpler linear models are less prone to overfitting when training samples are limited. In contrast, Random Forest (RF) benefits from the larger 3-sec dataset, where increased data diversity allows it to better capture complex non-linear feature interactions. The proposed Weighted Hybrid model consistently achieves the highest accuracy by combining the strong generalization of LR with the non-linear learning capability of RF, resulting in more robust and reliable predictions.

Effect of Dataset Size: Dataset granularity plays a critical role in model performance. The 30-second dataset contains fewer but longer samples, which limits the total number of training instances available to the model. Segmenting tracks into 3-second clips dramatically increases the effective dataset size and diversity, enabling models to learn richer local temporal and spectral patterns.

7 TASK 2. Dimensionality Reduction

To mitigate redundancy, we employ **Principal Component Analysis (PCA)**. PCA transforms the feature space into orthogonal components $Z = XW$. Standard scaling is performed before retaining **95% of total variance**.

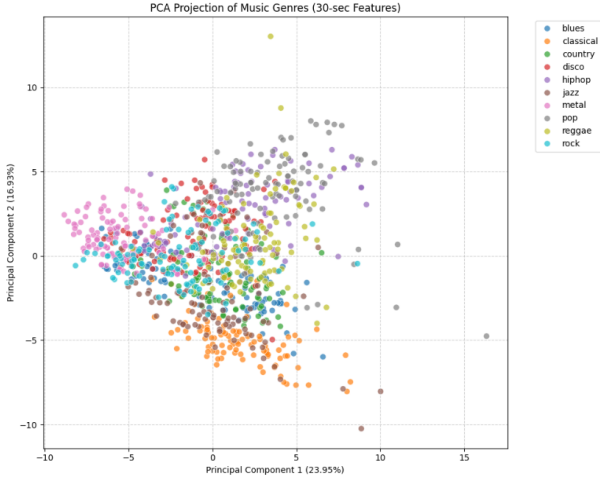


Figure 6: 2D PCA projection of music genres.

Table 1: Structured comparison of input representations and training configurations.

Configuration	Classical (30s)	Classical (3s)	Deep Learning	Wav2Vec2	MERT
Input Type	Features	Features	Spectrogram	Raw Audio	Raw Audio
Input Shape	(1, 60)	(1, 60)	(128,128,1)	—	—
Models	RF, LR	RF, LR	CNN / ResNet / LSTM	W2V2 + MLP	MERT + MLP
Feature Learning	Manual	Manual	Learned	Self-Supervised	Self-Supervised
Dimensionality	PCA (95%)	PCA (95%)	—	Embeddings	Embeddings
Batch Size	—	—	32	16	4
Optimizer	—	—	Adam / AdamW	AdamW	AdamW

Table 3: PCA Dimensionality Reduction Results

Dataset	Dimensions	Accuracy	Macro F1
30-sec	57 → 33	72.0%	0.719
3-sec	57 → 39	72.0%	0.716

Note: Approximately 40% dimensionality reduction was achieved with minimal performance degradation.

Insight: While PCA is useful for stabilizing models on limited data (improving 30-sec RF from 0.690 to 0.720), it acts as a bottleneck for the 3-sec configuration (dropping performance from 0.775 to 0.720) by discarding subtle spectral nuances vital for large-scale learning.

8 TASK 3. Spectrogram Deep Learning

We utilize precomputed spectrograms (128x128x1) to encode time-frequency characteristics.

8.1 Convolutional Neural Network (CNN)

Our custom CNN consists of four convolutional blocks (32, 64, 128, 256 channels) with Batch Normalization, ReLU, and MaxPooling. The classifier uses a dense layer (512), Dropout ($p = 0.4$), and a final 10-neuron output. Training uses Adam ($\text{lr}=0.0005$, $\text{batch}=32$). The baseline CNN achieved 68.0% accuracy.

8.2 Hybrid CNN–LSTM Model (Novelty)

We propose a hybrid CNN–LSTM architecture to simultaneously capture spatial and temporal characteristics from Mel-spectrograms.

8.2.1 Spatial and Temporal Feature Extraction

CNN Branch: Processes spectrograms as 2D inputs to extract frequency-time correlations and local patterns. Using convolutional and pooling layers, it generates a compact representation of high-level spectral features.

LSTM Branch: Models long-term dependencies by treating spectrogram frames as a temporal sequence. Bidirectional LSTM layers capture bidirectional context, effectively learning the rhythmic and structural dynamics of the audio.

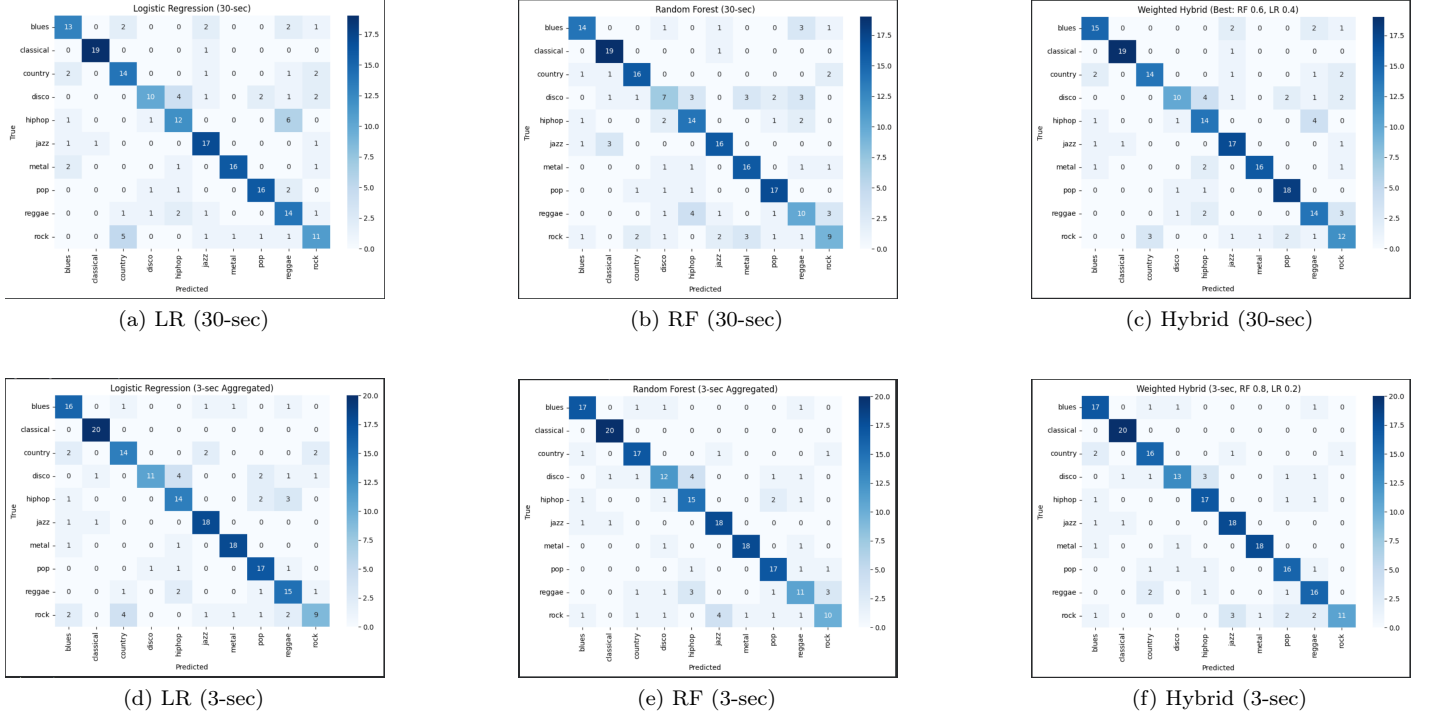


Figure 3: Confusion matrices of Logistic Regression (LR), Random Forest (RF), and Weighted Hybrid models on the 30-second and 3-second datasets.

8.2.2 Fusion and Classification

The spatial features from the CNN and temporal embeddings from the LSTM are concatenated into a unified vector. This fused representation is passed through fully connected layers for the final genre classification.

8.2.3 Key Insight

The CNN branch identifies *spectral patterns*, while the LSTM captures *temporal evolution*. Their integration enables a holistic understanding of music, significantly improving classification robustness.

9 Audio Transformers

We evaluate **Wav2Vec2** and **MERT** to analyze the impact of self-supervised representations on musical audio. The latent representation $\mathbf{z}_t$ is extracted via:

$$\mathbf{z}_t = \text{Transformer}(\text{CNN}(\text{Audio})) \quad (2)$$

Results and Discussion: Experimental results reveal a significant performance gap. **Wav2Vec2**, being speech-pretrained, achieved only **47.68%** accuracy as it focuses on phonetic formants rather than musical harmony. In contrast, **MERT** leverages massive-scale music pretraining to capture complex timbral and melodic structures, yielding a superior accuracy of **81.33%**. This highlights that domain-specific knowledge is critical for high-fidelity MIR tasks.

10 TASK 5. Comparative Analysis

Best Performing Architecture: The quantitative results identify **MERT (Music Transformer)** as the superior model (**81.33%** Accuracy), followed by the **Hybrid CNN-LSTM (78.15%)**. MERT’s dominance stems from massive-scale **domain-specific pretraining**, enabling it to understand complex musical grammar natively. Conversely, supervised models like CNN-LSTM are limited by the small GTZAN dataset size (1,000 samples).

Table 4: Performance Comparison Summary

Model	Data Type	Acc (%)
LR (30-sec)	Handcrafted	71.0
Weighted Hybrid (3-sec)	Handcrafted	81.0
Custom CNN	Mel-Spec	68.0
Transfer Learning (ResNet18)	Mel-Spec	74.83
Hybrid CNN-LSTM	Mel-Spec	78.15
Wav2Vec2	Raw Audio	47.68
MERT	Raw Audio	81.33

10.1 Architectural Trade-offs

- **Classical ML:** Establishes a moderate baseline. While **interpretable** and lightweight, it is bottlenecked by the expressive power of manually engineered features (MFCCs).
- **Deep Learning:** Provides significant accuracy gains by learning **hierarchical representations** from spectro-

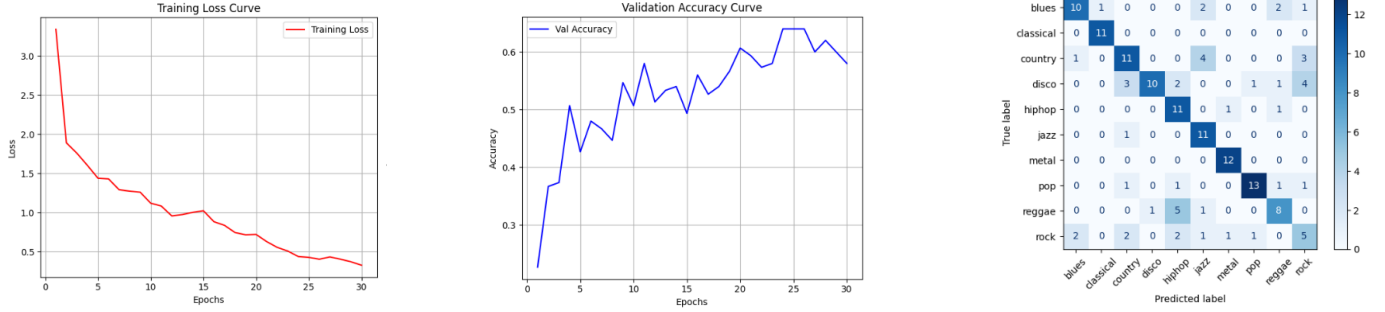


Figure 7: Training performance (Loss, Accuracy, and Confusion Matrix) of the self-designed CNN model.

grams, but requires high compute and careful regularization to prevent overfitting.

10.2 The Impact of Data Representation

This study reveals that **representation quality** is the primary performance driver:

- **1D Handcrafted Features:** Performance is capped by human intuition.
- **2D Spectrograms:** Yields a $\sim 7.15\%$ boost by allowing CNNs to capture harmonic edges and textures.
- **Semantic Embeddings:** Provides the highest accuracy by mapping audio into a rich, pretrained latent space.

Domain-Mismatch Penalty: Wav2Vec2 (47.68%) fails because speech-optimized models cannot interpret polyphonic music, proving the need for **music-specific pre-training**.

10.3 Limitations

- **Dataset Scale:** GTZAN’s small size increases overfitting risks for large transformers.
- **Segment Dependency:** 3-second clips may lack representative traits (e.g., quiet intros), leading to local misclassifications.

11 Music Recommendation System (Novelty)

Given two feature vectors $\mathbf{x}$ and $\mathbf{y}$, similarity is computed using cosine similarity:

$$\text{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} \quad (3)$$

Confidence Estimation: Reliability is estimated via maximum softmax probability ($\max p_i$), while recommendation strength is derived from raw cosine similarity scores.

By leveraging MERT embeddings, the engine resolves the **cold start** problem of collaborative filtering. It replaces historical user data with **acoustic topology**, matching the input’s tempo, polyphony, and timbre to similar tracks in the latent space.

12 Conclusion and Future Work

Conclusion: This study confirms that feature representation is the primary determinant of success in Music Genre Classification. While classical models provide a reliable baseline, deep learning architectures significantly elevate performance by learning hierarchical time-frequency patterns.

Our findings highlight a clear performance trajectory as data abstraction increases:

$$\text{Handcrafted (1D)} \xrightarrow{+7.15\%} \text{Spectrograms (2D)} \xrightarrow{+3.18\%} \text{Learned Embeddings}$$

Ultimately, while the Hybrid CNN–LSTM model effectively captures temporal dynamics, the domain-specific semantic depth of **MERT** remains the state-of-the-art for MIR tasks.

Future Work: Future research will focus on:

- **Late-Fusion Models:** Combining MFCCs with transformer embeddings to capture both timbral and structural features.
- **Self-Attention:** Implementing temporal attention to prioritize definitive segments (e.g., choruses) over noise.
- **Data Augmentation:** Using pitch-shifting and time-stretching to improve model robustness on diverse audio.

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